

Computer Communication Networks

Course Code: CSE248 | Spring 2026

GNS3 Network Simulation Project

Phase 1 — Setup & Proposal Report

Project Title: SOHO Network Design & GNS3 Simulation

Network Type: Small Office / Home Office (SOHO)

Group Name: *crackhead*

Student Name: Emad Yar Khan

ERP Number: 31651

Instructor: M. Zain Uddin

1. Selected Network Type

1.1 Network Type: Small Office / Home Office (SOHO)

The selected network type for this project is the Small Office / Home Office (SOHO) configuration. This is a single-member project and the SOHO option is the designated choice for individual submissions as outlined in the project brief.

1.2 Justification

The SOHO network type was selected for the following reasons:

- Relevance to real-world scenarios: SOHO networks are among the most widely deployed network types globally, found in homes, freelance offices, and small businesses. Understanding their design is a foundational networking skill.
- Appropriate complexity for individual work: The SOHO topology provides sufficient technical depth — including routing, DHCP, NAT, and firewall configuration — while remaining manageable for a single student within the four-week timeframe.
- Coverage of core CCN concepts: Implementing a SOHO network requires applying IP addressing, subnetting, dynamic host configuration, and network address translation — all core topics from the CSE248 curriculum.
- GNS3 compatibility: The required devices (one router, one switch, four PCs) are lightweight and run efficiently within GNS3 on a VirtualBox-hosted Ubuntu VM without excessive resource demands.

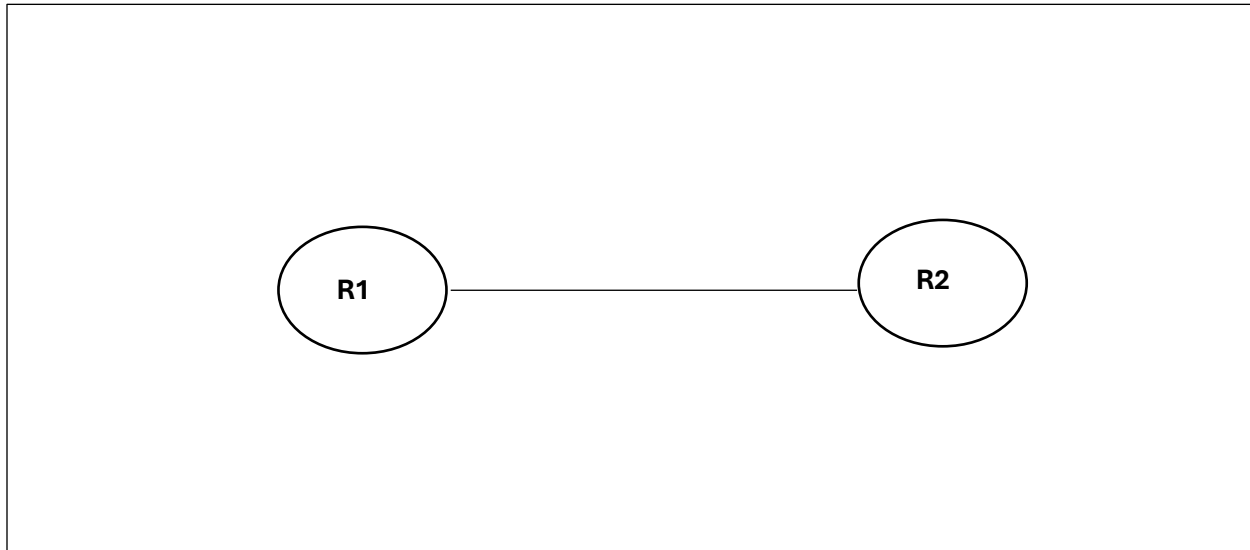
1.3 Minimum Requirements Compliance

Requirement	Status	Details
1 Router	Planned	Cisco C3725 in GNS3
1 Switch	Planned	Cisco IOS Switch (Layer 2)
4 PCs	Planned	GNS3 VPCS or Virtual PCs
DHCP	Planned	Configured on router fa0/0
NAT	Planned	PAT on WAN interface fa0/1
Basic Firewall Rules	Planned	ACLs on WAN-facing interface

2. Network Topology Diagram

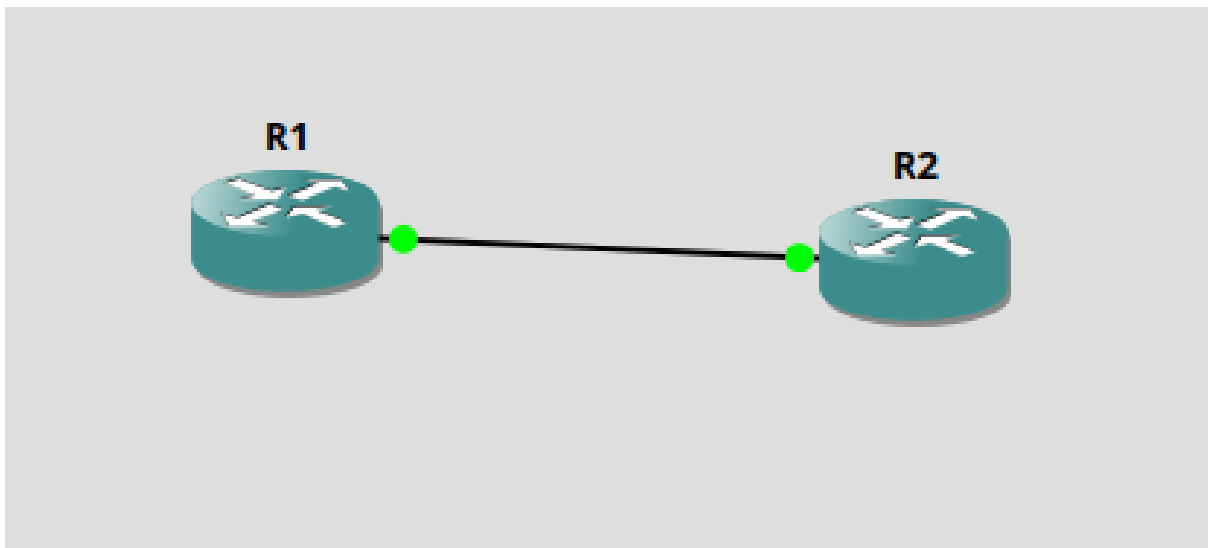
2.1 Logical Topology

The SOHO network consists of a single router acting as the gateway, connected to a managed switch which distributes connectivity to four end-user PCs. The router also connects to a simulated ISP/WAN interface to represent internet access. The logical layout is shown below:



2.2 Physical Topology

The physical topology mirrors the logical layout. All devices are simulated within GNS3 running on an Ubuntu 24.04 VM hosted in VirtualBox. No physical hardware is used.



2.3 Device List

Device	Hostname	Model	Role
Router	R1	Cisco C3725	Default gateway, DHCP server, NAT, ACL
Switch	SW1	Cisco IOS L2 Switch	Layer 2 switching for LAN devices
PC1	PC1	GNS3 VPCS	End-user workstation
PC2	PC2	GNS3 VPCS	End-user workstation
PC3	PC3	GNS3 VPCS	End-user workstation
PC4	PC4	GNS3 VPCS	End-user workstation

3. IP Addressing Plan

3.1 Subnet Design

The SOHO network uses a single /24 subnet for the LAN. A separate /30 subnet is used for the WAN link between the router and the simulated ISP. The addressing scheme is summarized below:

Segment	Network Address	Prefix	Usable Range	Purpose
LAN	192.168.1.0	/24	192.168.1.1 – .254	Internal SOHO LAN
WAN	203.0.113.0	/30	203.0.113.1 – .2	Router to ISP link

3.2 Device IP Address Table

Device	Interface	IP Address	Subnet Mask	Gateway	DHCP?
R1	fa0/0 (LAN)	192.168.1.1	255.255.255.0	—	No
R1	fa0/1 (WAN)	203.0.113.2	255.255.255.252	203.0.113.1	No
SW1	VLAN 1 (Mgmt)	192.168.1.2	255.255.255.0	192.168.1.1	No
PC1	eth0	192.168.1.10	255.255.255.0	192.168.1.1	Yes
PC2	eth0	192.168.1.11	255.255.255.0	192.168.1.1	Yes
PC3	eth0	192.168.1.12	255.255.255.0	192.168.1.1	Yes
PC4	eth0	192.168.1.13	255.255.255.0	192.168.1.1	Yes

3.3 DHCP Pool Configuration (Planned)

The router (R1) will serve as the DHCP server for the LAN. The planned DHCP pool configuration is as follows:

Parameter	Value
Pool Name	SOHO_POOL
Network	192.168.1.0 / 255.255.255.0
Default Gateway	192.168.1.1
DNS Server	8.8.8.8 (Google DNS)
Lease Duration	1 day (default)
Excluded Addresses	192.168.1.1 – 192.168.1.9 (reserved for infrastructure)
Dynamic Range	192.168.1.10 – 192.168.1.254

4. Services to be Implemented

The following network services and protocols will be configured during Weeks 2 and 3 of the project:

#	Service / Protocol	Description
1	DHCP	Dynamic Host Configuration Protocol — R1 will automatically assign IP addresses, subnet masks, default gateways, and DNS server addresses to all four PCs on the LAN.
2	NAT / PAT	Network Address Translation (Port Address Translation) — R1 will translate private LAN addresses (192.168.1.x) to the public WAN IP (203.0.113.2), enabling simulated internet access for all PCs through a single public IP.
3	Static Routing	A default route will be configured on R1 pointing to the simulated ISP gateway (203.0.113.1) for internet-bound traffic.
4	Access Control Lists (ACL)	Basic firewall rules using Standard or Extended ACLs will be applied on the WAN interface (fa0/1) to control inbound traffic and restrict unauthorized access to the LAN.
5	DNS (Client-side)	PCs will be configured with 8.8.8.8 as the DNS server via DHCP. No local DNS server will be hosted; resolution will be forwarded to the public DNS.

5. Division of Work

This project is completed individually. All tasks are the sole responsibility of the student listed on the title page. The work distribution across the four-week timeline is as follows:

Week	Phase	Tasks	Responsible
Week 1	Setup & Proposal	Install VirtualBox, Ubuntu VM, GNS3. Import Cisco IOS image. Build test topology. Write and submit Phase 1 proposal report.	[YOUR NAME]
Week 2	Design & Build	Build full SOHO topology in GNS3. Assign all IPs. Configure router interfaces, switch, and VPCS PCs. Set up DHCP and default route.	[YOUR NAME]
Week 3	Testing & Documentation	Configure NAT/PAT and ACLs. Test full connectivity. Run ping/traceroute tests. Capture screenshots. Begin writing final report.	[YOUR NAME]
Week 4	Final Submission	Finalize and polish final report. Record 10–15 min demo video with narration. Package GNS3 project files. Submit all deliverables.	[YOUR NAME]

6. References

The following sources have been consulted in the preparation of this proposal and will continue to be referenced throughout the project:

1. Cisco Systems, Inc. (2024). Cisco IOS IP Addressing Services Configuration Guide. Cisco Press. Available at: <https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/ipaddr/configuration/x-16/ipaddr-xe-16-book.html>
2. GNS3 Technologies Inc. (2024). GNS3 Documentation — Getting Started. Available at: <https://docs.gns3.com/docs/>
3. Hegde, P. (2024). *Cisco Images for GNS3 and EVE-NG* [GitHub Repository]. GitHub. Available at: <https://github.com/hegdepavankumar/Cisco-Images-for-GNS3-and-EVE-NG>

— End of Phase 1 Proposal Report —